

Keynote Speakers

IRI 2026

Keynote I

AI for Cyber and Cyber for AI



Bhavani Thuraisingham
The University of Texas at Dallas

Abstract

Artificial Intelligence (AI) and Cyber Security (Cyber) have been evolving for the past 60–75 years. AI was born when Turing published his famous paper in 1950 known as “the Imitation Game” and also called the “Turing Test.” It was essentially about the question “Can Machines Think?” Since then, AI continued to develop throughout the 1950s, 1960s and 1970s, including expert systems and machine learning systems.

At the same time, while modern computers were born in the 1940s, by the 1960s much progress was made in various topics in computing, including theory, systems and databases. By the late 1960s, computer security (known now as Cyber Security or Cyber) techniques such as access control were developed for operating systems and later for database systems. Both AI and Cyber continued to develop independently throughout the 1980s and 1990s, with some interaction between them.

With the growth of the World Wide Web in the 1990s, attacks on web-based systems exploded and various cyber security solutions were developed, including those that used AI. Then, with breakthroughs in machine learning through deep learning techniques in the early 2000s, progress in big data technologies around the same time, and more recently large language models (LLMs) since the late 2010s, AI has become a critical technology for every application, including Cyber.

However, AI techniques could themselves be attacked, rendering them useless. Therefore, in addition to AI being applied for cyber security solutions, cyber security solutions were also developed for AI systems in the 2010s and beyond. This presentation will examine several aspects of AI for Cyber and Cyber for AI, including security, privacy, integrity, dependability, secure

system design, governance, and hardware. We also discuss how quantum computing will revolutionize computing and be integrated with AI and Cyber.

Bio

Dr. Bhavani Thuraisingham is the Founders Chair Professor of Computer Science, the Founding Executive Director of the Cyber Security Research and Education Institute (CSI) at the University of Texas at Dallas (UTD; 2004–2021), and a senior strategist of the Trustworthy and Secure AI Center at UTD since 2023. She is an elected Fellow of the ACM, IEEE, AAAS, NAI, and IRI.

Her research interests are in integrating cyber security and artificial intelligence, including as they relate to the Cloud and Transportation Systems. She has received several technical, education and leadership awards, including the IEEE CS 1997 Edward J. McCluskey Technical Achievement Award, the IEEE CS 2023 Taylor L. Booth Education Award, the IEEE ComSoc Communications and Information Security 2019 Technical Recognition Award, the IEEE CS Technical Committee on Services Computing 2017 Research Innovation Award, the IEEE CS Technical Committee on Multimedia Computing Impact Award, the ACM SIGSAC 2010 Outstanding Contributions Award, the ACM CODASPY 2017 Lasting Research Award, the ACM SACMAT 10-Year Test of Time Awards in 2018 and 2019, and a 2013 IBM Faculty Award for Secure Cloud Computing.

Her 46-year career includes industry (Honeywell), a federal research laboratory (MITRE), U.S. government (NSF), and academia. Her work has resulted in 140+ journal articles, 300+ conference papers, 200+ keynote and featured addresses, seven U.S. patents, sixteen books, and more than 120 panel presentations, including at Fortune Media, Lloyd's of London Insurance, Dell Technologies World, the United Nations, and the White House Office of Science and Technology Policy.

She has also written opinion columns for venues such as The New York Times, Inc. Magazine, Womensday.com, and The Legal 500. She received her Ph.D. from the University of Wales, Swansea, UK, and the prestigious earned higher doctorate (D.Eng.) from the University of Bristol, UK. She also has a Certificate in Public Policy Analysis from the London School of Economics and Political Science. She was featured in the ACM's 2024 book "Rendering History: The Women of ACM-W" as one of the 30+ "Women that Changed the Face of World Wide Computing Forever."

Keynote Speaker II

Beyond Models: The Real Data Transformation Behind Generative AI



Roger Barga
University of Washington

Abstract

In this presentation, I will examine how generative AI is reshaping data infrastructure, analytical workflows, and enterprise decision-making. The discussion begins with the massive capital investment behind modern AI systems, the emerging commoditization of frontier models, and the practical realities of deployment across industries.

I will highlight where AI is already delivering measurable value—software development, automation, and customer operations—and how new analytical capabilities will unlock second-order effects similar to past platform shifts.

For data leaders, the talk offers a strategic lens on how probabilistic systems change the nature of data, how new forms of aggregation and unbundling will alter business models, and why AI's long-term impact will come from the questions it enables organizations to ask, not just the tasks it automates.

Bio

Roger Barga is a cloud and artificial intelligence executive whose career has been defined by building transformative technologies that bridge advanced research and real-world impact. With over four decades of experience spanning academia, research, and executive leadership at Oracle, Microsoft, and Amazon Web Services, he has consistently operated at the intersection of innovation and enterprise execution.

His work reflects a deep commitment to helping organizations harness the power of data, cloud, and machine learning to solve meaningful, complex problems at scale. Roger holds a Ph.D. in Computer Science and serves as a lecturer in Data Science and Machine Learning at the University of Washington.

Keynote Speaker III

Can Artificial Intelligence Solve Longstanding Challenges in Information Reuse and Integration in Public-Private Partnerships?



John Y. Choe
University of Washington, Seattle

Abstract

Artificial intelligence offers powerful new tools for information reuse and integration, including metadata generation, knowledge graphs, data harmonization, privacy-preserving computation, and decision support. Yet in public-private partnerships, especially those supporting disaster resilience and emergency preparedness, the most persistent barriers are not technical alone.

Relevant data are distributed across agencies, firms, healthcare systems, infrastructure operators, community organizations, researchers, and local partners, each with different formats, authorities, risks, incentives, and trust relationships.

This keynote asks whether AI can solve these longstanding challenges. I argue that AI can help, but only when embedded in trustworthy socio-technical data ecosystems. Without provenance, governance, privacy protections, contextual metadata, and mutual benefit, AI may make fragmented information appear more coherent than it is.

Using disaster resilience and emergency preparedness as motivating domains, the talk connects regional data ecosystem design to core IRI themes: information reuse, information integration, reusable systems, data lineage, knowledge graphs, ontologies, harmonization, hallucination mitigation, and AI-enabled decision support. The keynote concludes with a research agenda for building AI systems that make not only data reusable, but also trust, accountability, and emergency decision workflows.

Bio

John Y. Choe, Ph.D., is an Associate Professor of Industrial & Systems Engineering at the University of Washington, Seattle, where he directs the Disaster Data Science Lab and serves as Deputy Director of the Center for Disaster Resilient Communities. His interdisciplinary research develops data science-grounded AI and decision-support systems to strengthen community resilience to extreme hazards.

His recent work focuses on disaster data ecosystems, public health emergency preparedness, and socio-technical approaches to information reuse and integration across public, private, and community partners. His work has been supported by the U.S. National Science Foundation, the National Institute of Environmental Health Sciences, the Centers for Disease Control and Prevention, and other public and private organizations.